



M1.6 Unit Overview

Rigid Transformations and Congruence

Students construct geometric figures using a variety of tools. They then use their construction strategies to perform rigid transformations. Students use the definition of congruence in terms of rigid transformations to establish that corresponding parts of congruent triangles are congruent. They then explain how the criteria for triangle congruence (ASA, SAS, and SSS) follow from this.

Let us know what you thought about this lesson by filling out [this survey](#).

Big Ideas: CC4: Shapes in Structures, Transformations and Congruence

Explore: Making Precise Symbols

(Optional) Launch the unit with a non-routine task to investigate the question, "How are geometric shapes and concepts used in art and culture?"

Sub-Unit 1: Constructions (Lessons 1–7 + Practice Day)

- Use a variety of construction tools (e.g., compass, straightedge, and dynamic geometry software) to construct precise geometric figures.
- Language Goal:** Write and analyze descriptions of constructions.

Sub-Unit 2: Transformations (Lessons 8–16 + Practice Day + Quiz)

- Perform and describe a sequence of rigid transformations that moves a figure onto another.
- Use coordinate notation to describe transformations.
- Language Goal:** Describe the relationships that exist when a figure is transformed by rigid transformations.

Sub-Unit 3: Congruence (Lessons 17–22 + Practice Day)

- Identify corresponding parts in diagrams and from congruence statements.
- Use rigid transformations to show SAS, ASA, and SSS triangle congruence theorems are true.
- Language Goals:**
 - Write a congruence statement using words or geometric symbols.
 - Explain SAS, ASA, and SSS triangle congruence using rigid transformations.

Review Vocabulary

center of rotation, clockwise, congruent, conjecture, converse, corresponding parts, counterclockwise, dilation, image, line segment, parallel line(s), perpendicular, pre-image, ray, reflection, rigid transformation, rotation, square, translation, vertex

Contextual Vocabulary

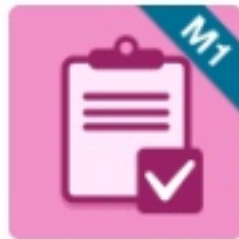
Australian Aboriginal, dot painting, hospital, library, region, Islamic cultures, underlying constructions, kaleidoscope

Standards

Addressing G-CO.1 G-CO.12 G-CO.13 G-CO.2 G-CO.3 G-CO.4 G-CO.5 G-CO.6 G-CO.7 G-CO.8 SMP.1 SMP.3 SMP.4 SMP.5 SMP.6 SMP.7 SMP.8

Vocabulary

- angle
- angle-side-angle triangle congruence theorem
- circle
- congruence statement
- construction
- equilateral triangle
- equidistant
- if and only if
- inscribe
- line of reflection
- line of symmetry
- midpoint (of a line segment)
- perpendicular bisector
- prime notation
- reflectional symmetry
- regular polygon
- rotational symmetry
- symmetry
- side-angle-side triangle congruence theorem
- side-side-side triangle congruence theorem
- Voronoi diagram



M1.6 Pre-Unit Check



Learn more about your students' understanding of foundational concepts and skills that will support them in the upcoming unit. This pre-unit check is intended for students to complete without a calculator.

OPTIONAL This is optional and can be given in its entirety before the unit begins or spread out throughout the unit.

Let us know what you thought about this lesson by filling out [this survey](#).

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC4: Shapes and Symmetries (Grade 4), Shapes in the World (Grade 7), Transformational Geometry (Grade 8)

Standards

Assessing 4.G.1 4.G.2 7.G.4 8.G.1 8.G.2 8.G.3 SMP.6 SMP.7



Explore: Making Precise Symbols

How are geometric shapes and concepts used in art and culture?

Let us know what you thought about this lesson by filling out [this survey](#).



Student Edition pages and Presentation Screens support learning in this lesson.

Big Idea: CC4: Transformations and Congruence

Today's Goals

- Draw symbols and geometric shapes or concepts with precision.
- Language Goal:** Describe what it means for a geometric drawing to be precise. (*Writing*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with the **Building Math Habits of Mind** part of this Explore to support math language acquisition for your students.

Additional Vocabulary

equilateral triangle

Additional Materials

straightedges, string, reflective devices (optional), scissors (optional)

Standards

Building On N-Q.3 Addressing G-CO.12 SMP.5 SMP.6 Building Toward G-CO.12 G-CO.13

Language Standards

ELD.PI.9-10.11 ELD.PI.9-10.2 ELD.PI.9-10.3

Review Vocabulary

- circle
- parallel lines
- perpendicular
- square

Instructional Routines

- Notice and Wonder
- Discussion Supports (MLR8)



Circles and Segments

Lesson 1: An Introduction to Constructions

Let's construct designs using circles and line segments.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use a digital circle and line tool to develop conceptual understanding of geometric constructions.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC4: Transformations and Congruence

Today's Goals

- Use a circle to construct different segments of the same length.
- Language Goal:** Explain how circles help to construct figures with equal side lengths. (*Reading, Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Additional Materials

Distance Around an Arc Activity (one per student)(optional)

Standards

Building On G-CO.1 Addressing G-CO.1 G-CO.12 SMP.7 SMP.8 Building Toward G-CO.12 G-CO.13

Language Standards

ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.2 ELD.PI.9-10.3 ELD.PI.9-10.6

Vocabulary

- circle
- construction

Review Vocabulary

- line segment

Instructional Routines

- Which One Doesn't Belong?



Compass Constructions

Lesson 2: Constructing Patterns

Let's use a compass to make precise designs on paper.

Let us know what you thought about this lesson by filling out [this survey](#).



This lesson uses print **student pages** and digital **Teacher Presentation Screens**.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC4: Transformations and Congruence

Today's Goals

- Use a compass and straightedge to construct *equilateral triangles* and *regular* hexagons.
- **Language Goal:** Describe processes for performing compass and straightedge constructions. (*Reading, Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Additional Vocabulary

compass, straightedge

Additional Materials

compasses; straightedges

Standards

Building On 7.G.2 Addressing G-CO.12 G-CO.13 SMP.6 SMP.6 SMP.7 SMP.7 Building Toward G-CO.12 G-CO.13

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.12 ELD.PI.9-10.3

Vocabulary

- equilateral triangle
- inscribe
- regular polygon

Instructional Routines

- Stronger and Clearer Each Time (MLR1)
- Compare and Connect (MLR7)



Constructing Digitally

Lesson 3: Geometric Constructions

Let's explore how to communicate to each other about constructions.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use the digital construction tools to make geometric constructions and justify why they are valid.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC4: Transformations and Congruence

Today's goals

- Use digital construction tools to construct geometric objects.
- Language Goal:** Write and follow precise instructions to create geometric objects. (*Reading and Writing*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Standards

Building On G-CO.12 Addressing G-CO.1 G-CO.12 G-CO.13 SMP.3 SMP.3 SMP.6 SMP.6 Building Toward G-CO.12

Language Standards

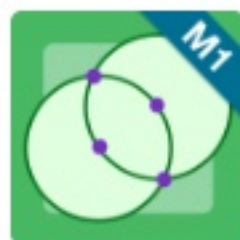
ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.3 ELD.PI.9-10.6

Vocabulary

- angle

Instructional Routines

- Stronger and Clearer Each Time (MLR1)



Right in the Middle

Lesson 4: Constructing Perpendicular Bisectors

Let's explore equal distances through constructions.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use digital construction tools to construct and analyze perpendicular bisectors and use their constructions to make generalizations.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC4: Transformations and Congruence

Today's Goals

- Construct the *perpendicular bisector* of a line segment.
- Language Goal:** Describe that points that are *equidistant* from a line segment's endpoints will be on the *perpendicular bisector* of that segment. (*Writing and Speaking*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Standards

Building On 7.G.2 Addressing G-CO.12 SMP.1 SMP.6 SMP.8 Building Toward G-CO.12 G-CO.13

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.3 ELD.PI.9-10.6

Vocabulary

- equidistant
- midpoint
- perpendicular bisector

Instructional Routines

- Compare and Connect (MLR7)

Lines of Construction

Lesson 5: Constructing Perpendicular and Parallel Lines

Let's develop strategies to construct perpendicular and parallel lines.

Let us know what you thought about this lesson by filling out [this survey](#).



This lesson uses **digital student screens** and print **student pages**. Print Student Notes pages are also available.

Why digital? Students use Responsive Feedback to determine whether their digital constructions are valid.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC4: Transformations and Congruence

Today's Goals

- Construct perpendicular and parallel lines that pass through given points.
- Language Goal:** Write precise instructions for completing a construction. (*Reading and Writing*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Additional Materials

compasses; straightedges

Standards

Building On 8.G.5 G-CO.12 Addressing G-CO.12 SMP.3 SMP.3 SMP.6 SMP.6 Building Toward G-CO.12 G-CO.13

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.3

Review Vocabulary

- parallel lines
- perpendicular

Instructional Routines

- Stronger and Clearer Each Time (MLR1)



Square Up

Lesson 6: Constructing Squares

Let's make squares using circles.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use the digital construction tools to quickly explore the relationship between inscribed squares and circles.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC4: Transformations and Congruence

Today's goals

- Construct a square and a square inscribed in a circle.
- Language Goal:** Describe how construction techniques, like determining a midpoint or a perpendicular line, are used to construct a square. (*Speaking and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Vocabulary

diagonal

Standards

Building On G-CO.1 G-CO.12 Addressing G-CO.12 G-CO.13 SMP.1 SMP.3 SMP.6 Building Toward G-CO.13

Language Standards

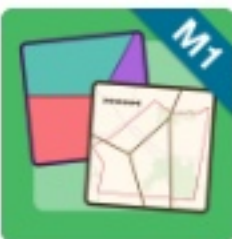
ELD.PI.9-10.1 ELD.PI.9-10.11 ELD.PI.9-10.3 ELD.PII.9-10.4

Review Vocabulary

- vertex

Instructional Routines

- Compare and Connect (MLR7)
- Notice and Wonder



Community Constructions

Lesson 7: Using Perpendicular Bisectors

Let's use constructions to help us make decisions.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students can use digital construction tools to create perpendicular bisectors and Voronoi diagrams.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC4: Transformations and Congruence

Today's Goals

- Construct perpendicular bisectors to create regions closest to a certain location.
- Language Goal:** Explain how perpendicular bisectors and Voronoi diagrams can help us make sense of where to put community resources. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Additional Materials

computers/devices (for research) (optional)

Standards

Building On G-CO.12 Addressing G-CO.12 SMP.1 SMP.3 SMP.4 SMP.6 Building Toward G-CO.4 G-MG.3

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.11 ELD.PI.9-10.2 ELD.PI.9-10.3 ELD.PI.9-10.6

Vocabulary

- Voronoi diagram

Instructional Routines

- Compare and Connect (MLR7)
- Notice and Wonder



M1.6 Practice Day 1

Students practice the concepts, skills, and strategies developed in Lessons 1–7. Consider using this Practice Day as preparation for the upcoming Sub-Unit Quiz.

Let us know what you thought about this lesson by filling out [this survey](#).

Big Idea: CC4: Transformations and Congruence

Materials

Option 1: Partner Practice

- Student Edition
- Task Cards (one per pair, pre-cut)

Option 2: Solve and Swap

- Student Edition
- Task Cards (one per student, pre-cut)

Standards

Addressing

G-CO.1

G-CO.12

G-CO.13

SMP.5

SMP.5



Going Off the Grid, Part 1

Lesson 8: Introducing Sequences of Transformations

Let's explore how we can describe sliding, flipping, and spinning shapes.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use digital tracing paper to make sense of and perform rigid transformations.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC4: Transformations and Congruence, Shapes in Structures

Today's Goals

- Design a sequence of rigid transformations that takes a figure onto another.
- Language Goal:** Write and follow instructions to perform a sequence of transformations. (*Reading, Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Standards

Building On 8.G.1 G-CO.1 Addressing G-CO.2 G-CO.5 G-CO.6 SMP.6 SMP.6 SMP.7 SMP.7

Building Toward G-CO.4 G-CO.6

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.6 ELD.PII.9-10.3 ELD.PII.9-10.4 ELD.PII.9-10.5

Vocabulary

- prime notation

Review Vocabulary

- clockwise
- counterclockwise
- image
- pre-image
- reflection
- rigid transformation
- rotation
- translation

Instructional Routines

- Stronger and Clearer Each Time (MLR1)



Time to Reflect

Lesson 9: Defining Reflections

Let's get precise about reflections.

Let us know what you thought about this lesson by filling out [this survey](#).



This lesson uses print **student pages** and digital **Teacher Presentation Screens**.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC4: Transformations and Congruence, Shapes in Structures

Today's Goals

- Reflect figures precisely using a variety of tools.
- **Language Goal:** Describe what happens to a *pre-image* when it is reflected over a given line. (*Speaking and Writing*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 3 to support math language acquisition for your students.

Additional Materials

compasses; straightedges; tracing paper; coloring tools

Standards

Building On 8.G.3 Addressing G-CO.2 G-CO.4 G-CO.5 G-CO.6 SMP.5 SMP.5 SMP.6 SMP.6 Building Toward G-CO.5 G-CO.6

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.5

Vocabulary

- line of reflection

Instructional Routines

- Think-Pair-Share



Translation Information

Lesson 10: Defining Translations

Let's get precise about translations.

Let us know what you thought about this lesson by filling out [this survey](#).



This lesson uses print **student pages** and digital **Teacher Presentation Screens**.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC4: Transformations and Congruence, Shapes in Structures

Today's Goals

- Translate figures precisely using a variety of tools.
- Language Goal:** Describe and interpret translations and reflections precisely. (*Speaking and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Additional Materials

compasses, straightedges, tracing paper, Activity 2 Cards (one set per pair, pre-cut), Show What You Know Sheet

Standards

Building On 8.G.1 Addressing G-CO.2 G-CO.4 G-CO.5 G-CO.6 SMP.5 SMP.5 SMP.6 SMP.6 Building Toward G-CO.5 G-CO.6

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.6

Instructional Routines

- Which One Doesn't Belong?



Rotation Devices

Lesson 11: Defining Rotations

Let's get precise about rotations.

Let us know what you thought about this lesson by filling out [this survey](#).

  This lesson uses print **student pages** and digital **Teacher Presentation Screens**.

Instruction Differentiation Beyond the Lesson

Big Ideas: CC4: Transformations and Congruence, Shapes in Structures

Today's Goals

- Rotate figures precisely using a variety of tools.
- **Language Goal:** Describe rotations using a center of rotation, direction of rotation, and angle of rotation. (*Writing and Speaking*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

compasses; straightedges; tracing paper; coloring tools

Standards

Building On 8.G.1 Addressing G-CO.2 G-CO.4 G-CO.5 G-CO.6 SMP.5 SMP.5 SMP.6 SMP.6

Building Toward G-CO.5 G-CO.6

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.6

Review Vocabulary

- center of rotation
- counterclockwise
- clockwise
- rotation

Instructional Routines

- Stronger and Clearer Each Time (MLR1)



Going Off the Grid, Part 2

Lesson 12: Specifying Sequences of Transformations

Let's use geometry tools to perform sequences of rigid transformations that move one figure onto another.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students receive Responsive Feedback as they explore a sequence of transformations that will move one figure onto another.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC4: Transformations and Congruence, Shapes in Structures

Today's Goals

- Use geometry tools to perform a sequence of rigid transformations that moves a figure onto another.
- Language Goal:** Describe a sequence of rigid transformations that moves one figure onto another using mathematically precise language. (*Writing and Speaking*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Additional Materials

tracing paper, coloring tools

Standards

Building On 8.G.2 G-CO.4 Addressing G-CO.2 G-CO.5 G-CO.6 SMP.5 SMP.5 SMP.6 SMP.6 SMP.7 SMP.7

Building Toward G-CO.3 G-CO.6

Language Standards

ELD.PI.9-10.10 ELD.PI.9-10.6

Instructional Routines

- Compare and Connect (MLR7)
- Stronger and Clearer Each Time (MLR1)



Tiles and Patterns

Lesson 13: Geometric Art Using Constructions and Transformations

Let's use transformations to describe and create patterns.

OPTIONAL This lesson is optional as the standards are addressed in other lessons within this course.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students are given an underlying construction and use intersection points to replicate a design. They use Responsive Feedback to help them make revisions when angles or line segments are slightly off.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC4: Transformations and Congruence, Shapes in Structures

Instructional Routines

- Notice and Wonder

Today's Goals

- Use an underlying construction to create polygons that can be transformed into geometric patterns.
- **Language Goal:** Describe connections between geometric patterns from Islamic cultures, constructions, and transformations. (*Reading, Writing, and Speaking*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

compasses, straightedges, tracing paper

Standards

Building On 7.G.2 G-CO.4 Addressing G-CO.12 G-CO.2 G-CO.5 G-CO.6 SMP.7 SMP.7 Building Toward G-CO.6

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.6 ELD.PII.9-10.4



Reflectional Symmetry

Lesson 14: Describing Reflections that Take a Figure Onto Itself

Let's explore what it means for a figure to have reflectional symmetry.

Let us know what you thought about this lesson by filling out [this survey](#).



Student Edition pages and Presentation Screens support learning in this lesson.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC4: Transformations and Congruence, Shapes in Structures

Today's Goals

- Determine whether a figure has reflectional symmetry.
- Determine the lines of symmetry for a figure.
- Language Goal:** Describe the reflections that take a figure onto itself. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

Activity 1 Sheet (one per student), scissors, coloring tools (optional), tracing paper, straightedges, compasses, Show What You Know Sheet

Standards

Building On 8.G.1 G-CO.4 Addressing G-CO.2 G-CO.3 G-CO.5 G-CO.6 SMP.1 SMP.6 SMP.7 Building Toward G-CO.3

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.12 ELD.PI.9-10.5

Vocabulary

- line of symmetry
- reflectional symmetry
- symmetry

Instructional Routines

- Notice and Wonder



Rotational Symmetry

Lesson 15: Describing Rotations that Take a Figure Onto Itself

Let's explore what it means for a figure to have rotational symmetry.

Let us know what you thought about this lesson by filling out [this survey](#).



Student Edition pages and Presentation Screens support learning in this lesson.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC4: Transformations and Congruence

Today's Goals

- Determine whether a figure has *rotational symmetry*.
- Sketch rotations that create symmetry.
- Language Goal:** Describe rotations that take a figure onto itself. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

tracing paper, compasses (optional), straightedges (optional), protractors (as needed)

Standards

Building On G-CO.3 G-CO.4 Addressing G-CO.2 G-CO.3 SMP.5 SMP.6 SMP.7

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.12 ELD.PI.9-10.3 ELD.PI.9-10.5

Vocabulary

- rotational symmetry

Review Vocabulary

- regular polygon

Instructional Routines

- Which One Doesn't Belong?
- Compare and Connect (MLR7)



Transformations as Functions

Lesson 16: Using Coordinate Transformation Notation

Let's explore how we can represent transformations as functions.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use Responsive Feedback to make connections between coordinate transformation notation and images created by different transformations.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC4: Transformations and Congruence

Today's Goals

- Use coordinate transformation notation as a function to take points in the plane as inputs and give transformed points as outputs.
- Language Goal:** Precisely define a transformation when given in coordinate transformation notation. (*Reading, Writing, and Speaking*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Additional Vocabulary

coordinate transformation notation

Standards

Building On G-CO.2 G-CO.5 Addressing G-CO.2 G-CO.5 SMP.3 SMP.6 SMP.7 Building Toward G-CO.6

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.3

Review Vocabulary

- dilation

Instructional Routines

- Decide and Defend
- Notice and Wonder



M1.6 Practice Day 2



Students practice the concepts, skills, and strategies developed in Lessons 1–16. Consider using this Practice Day as preparation for the upcoming End-of-Unit Assessment.

Let us know what you thought about this lesson by filling out [this survey](#).

Big Ideas: CC4: Transformations and Congruence, Shapes in Structures

Materials

Option 1: Stations

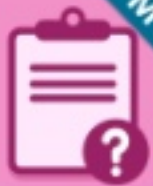
- Student Edition
- Digital Companion
- Task Cards (one per station, pre-cut)
- compasses, straightedges, tracing paper

Option 2: Partner Practice

- Student Edition
- Digital Companion
- Task Cards (one per pair, pre-cut)
- compasses, straightedges, tracing paper

Standards

Addressing G-CO.12 G-CO.2 G-CO.3 G-CO.5 G-CO.6 SMP.5 SMP.5 SMP.6 SMP.6



M1.6 Sub-Unit Quiz

Learn about your students' understanding of the concepts and skills so far in this unit.
This assessment is intended for students to complete without a calculator.

Let us know what you thought about this lesson by filling out [this survey](#).

Instruction

Differentiation Beyond the Lesson

Big Idea: CC4: Transformations and Congruence

Standards

Assessing

G-CO.1

G-CO.12

G-CO.4

G-CO.5

SMP.5



Congruent Parts, Part 1

Lesson 17: Defining Congruence

Let's inspect what corresponding sides and angles in figures have to do with congruence.

Let us know what you thought about this lesson by filling out [this survey](#).

 Student Edition pages and an assignable Digital Companion needed.

Instruction Differentiation Beyond the Lesson

Big Ideas: CC4: Shapes in Structures, Transformations and Congruence

Today's Goals

- Identify corresponding parts in diagrams.
- Language Goal:** Explain why figures and corresponding parts are congruent using rigid transformations. (*Reading, Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

tracing paper, coloring tools (optional)

Standards

Building On 8.G.1 8.G.2 G-CO.2 Addressing G-CO.5 G-CO.6 G-CO.7 SMP.3 SMP.5 SMP.6 Building Toward G-CO.7 G-CO.8

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.3 ELD.PI.9-10.6

Review Vocabulary

- congruent
- corresponding parts
- rigid transformation

Instructional Routines

- Notice and Wonder
- Decide and Defend



Congruent Parts, Part 2

Lesson 18: Congruence Statements

Let's name figures in ways that help us see corresponding parts.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use digital interactions to visualize corresponding parts and make sense of congruence statements.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC4: Shapes in Structures

Today's Goals

- Identify corresponding parts from a *congruence statement*.
- Language Goal:** Write a congruence statement using words or geometric symbols. (*Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

tracing paper

Standards

Building On G-CO.2 G-CO.5 G-CO.6 Addressing G-CO.6 SMP.1 SMP.6 SMP.7 Building Toward G-CO.7 G-CO.8

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.12 ELD.PI.9-10.3 ELD.PI.9-10.6

Vocabulary

- congruence statement

Review Vocabulary

- congruent
- corresponding parts
- rigid transformation

Instructional Routines

- Notice and Wonder
- Collect and Display (MLR2)



Congruent Triangles, Part 1

Lesson 19: Determining Information Needed for Triangle Congruence

Let's use rigid transformations to determine whether two triangles are congruent.

Let us know what you thought about this lesson by filling out [this survey](#).



This lesson uses print **student pages** and digital **Teacher Presentation Screens**.

Instruction

Differentiation Beyond the Lesson

Big Idea: CC4: Shapes in Structures

Today's Goals

- Show congruence of two triangles whose corresponding parts are congruent by determining and performing rigid transformations.
- Language Goal:** Describe why parts of two congruent triangles coincide using properties of rigid transformations. (*Reading, Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 2 to support math language acquisition for your students.

Additional Vocabulary

coincide

Standards

Building On 8.G.1.a 8.G.1.b G-CO.5 G-CO.6 Addressing G-CO.6 G-CO.7 SMP.3 SMP.6 Building Toward G-CO.8

Language Standards

ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.5 ELD.PII.9-10.5

Vocabulary

- if and only if

Review Vocabulary

- converse
- ray
- reflection
- rigid transformation
- rotation
- translation

Instructional Routines

- Stronger and Clearer Each Time (MLR1)



Congruent Triangles, Part 2

Lesson 20: Determining Shortcuts for Triangle Congruence

Let's explore shortcuts for determining whether two triangles are congruent.

Let us know what you thought about this lesson by filling out [this survey](#).

  Student Edition pages and Presentation Screens support learning in this lesson.

Instruction Differentiation Beyond the Lesson

Big Idea: CC4: Shapes in Structures

Today's Goals

- Create congruent triangles given angle measures and/or side lengths.
- Language Goal:** Determine and request the specific information needed to create congruent triangles. (*Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

Activity 1 Cards (one set per pair, pre-cut), coloring tools, protractors, rulers, Show What You Know Sheet

Standards

Building On 7.G.2 G-CO.7 Addressing SMP.1 SMP.3 SMP.6 Building Toward G-CO.8

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.2 ELD.PI.9-10.3 ELD.PI.9-10.5 ELD.PI.9-10.6

Review Vocabulary

- congruent
- corresponding parts

Instructional Routines

- Compare and Connect (MLR7)



Triangle Palindromes

Lesson 21: ASA and SAS Triangle Congruence Theorems

Let's explore two ways to show triangle congruence.

Let us know what you thought about this lesson by filling out [this survey](#).

  Student Edition pages and Presentation Screens support learning in this lesson.

Instruction Differentiation Beyond the Lesson

Big Idea: CC4: Shapes in Structures

Today's Goals

- Use rigid transformations to show SAS and ASA triangle congruence theorems are true.
- Language Goal:** Use SAS or ASA triangle congruence to explain why two triangles are congruent. (*Reading, Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials

protractors, rulers, coloring tools (as needed), tracing paper (as needed), *To Be Rigid* Activity Sheet (one per student) (optional)

Note: Consider using the *To Be Rigid* Activity after this lesson to support student understanding of the rigidity of triangles.

Standards

Building On G-CO.6 G-CO.7 Addressing G-CO.7 G-CO.8 SMP.1 SMP.3 SMP.6 Building Toward G-SRT.5

Language Standards

ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.11 ELD.PI.9-10.12 ELD.PI.9-10.6

Vocabulary

- angle-side-angle triangle congruence theorem
- side-angle-side triangle congruence theorem

Review Vocabulary

- conjecture

Instructional Routines

- Compare and Connect (MLR7)



Another Triangle Congruence Theorem

Lesson 22: Side-Side-Side Triangle Congruence Theorem

Let's explore one more congruence theorem.

Let us know what you thought about this lesson by filling out [this survey](#).



Student devices recommended. Student Edition pages are also available.

Why digital? Students use digital manipulatives to develop conceptual understanding of triangle congruence.

Instruction

Differentiation Beyond the Lesson

Big Ideas: CC4: Shapes in Structures, Transformations and Congruence

Today's Goals

- Determine if two triangles are congruent by SAS, ASA, or SSS congruence.
- Language Goal:** Use rigid transformations to explain SSS triangle congruence. (*Reading, Writing, Speaking, and Listening*)

Math Language Development

Consider using the scaffolds from the *Math Language Development Resources* with Activity 1 to support math language acquisition for your students.

Additional Materials: coloring tools (as needed), scientific calculators (as needed)

Standards

Building On G-CO.6 G-CO.7 Addressing G-CO.5 G-CO.8 SMP.1 SMP.3 SMP.6 Building Toward G-SRT.5

Language Standards

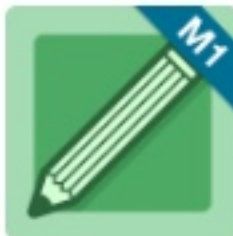
ELD.PI.9-10.1 ELD.PI.9-10.10 ELD.PI.9-10.3 ELD.PI.9-10.5 ELD.PI.9-10.6

Vocabulary

- side-side-side triangle congruence theorem

Instructional Routines

- Critique, Correct, Clarify (MLR3)



M1.6 Practice Day 3



Students practice the concepts, skills, and strategies developed in Lessons 1–22. Consider using this Practice Day as preparation for the upcoming End-of-Unit Assessment.

Let us know what you thought about this lesson by filling out [this survey](#).

Big Ideas: CC4: Transformations and Congruence, Shapes in Structures

Materials

Option 1: Solve and Swap

- Student Edition
- Problem Cards (one copy per eight students, pre-cut)
- compasses, straightedges

Option 2: Partner Practice

- Student Edition
- Problem Cards (one set per pair, pre-cut)
- compasses, straightedges

Standards

Addressing G-CO.12 G-CO.2 G-CO.3 G-CO.5 G-CO.6 G-CO.8 SMP.3 SMP.5 SMP.6



M1.6 Performance Task



Assign this summative assessment performance task at the end of the unit to evaluate students' proficiency with the concepts and skills addressed in the unit.

Let us know what you thought about this lesson by filling out [this survey](#).

Big Ideas: CC4: Shapes in Structures, Transformations and Congruence

Standards

Assessing G-CO.12 G-CO.2 G-CO.3 G-CO.5 G-CO.6 G-CO.8 SMP.3 SMP.5 SMP.7



Unit Synthesis and Reflection



Here are six optional activities for students to engage in at the end of a unit to synthesize and/or reflect on their learning from the unit.

Notes:

- **Select one to two tasks** for students to complete before or after the End-Unit Assessment.
- Many tasks invite students to refer to the Student Goals and Glossary, which can be downloaded from the Unit Overview page at the beginning of each unit.

🗨️ Let us know what you thought about this lesson by filling out [this survey](#).

Learning Goals

- Be purposeful, strategic, and goal-oriented about learning.
- Connect different ideas from the unit and across units.
- Revisit ideas from the unit in order to make revisions and reflect on their growth.